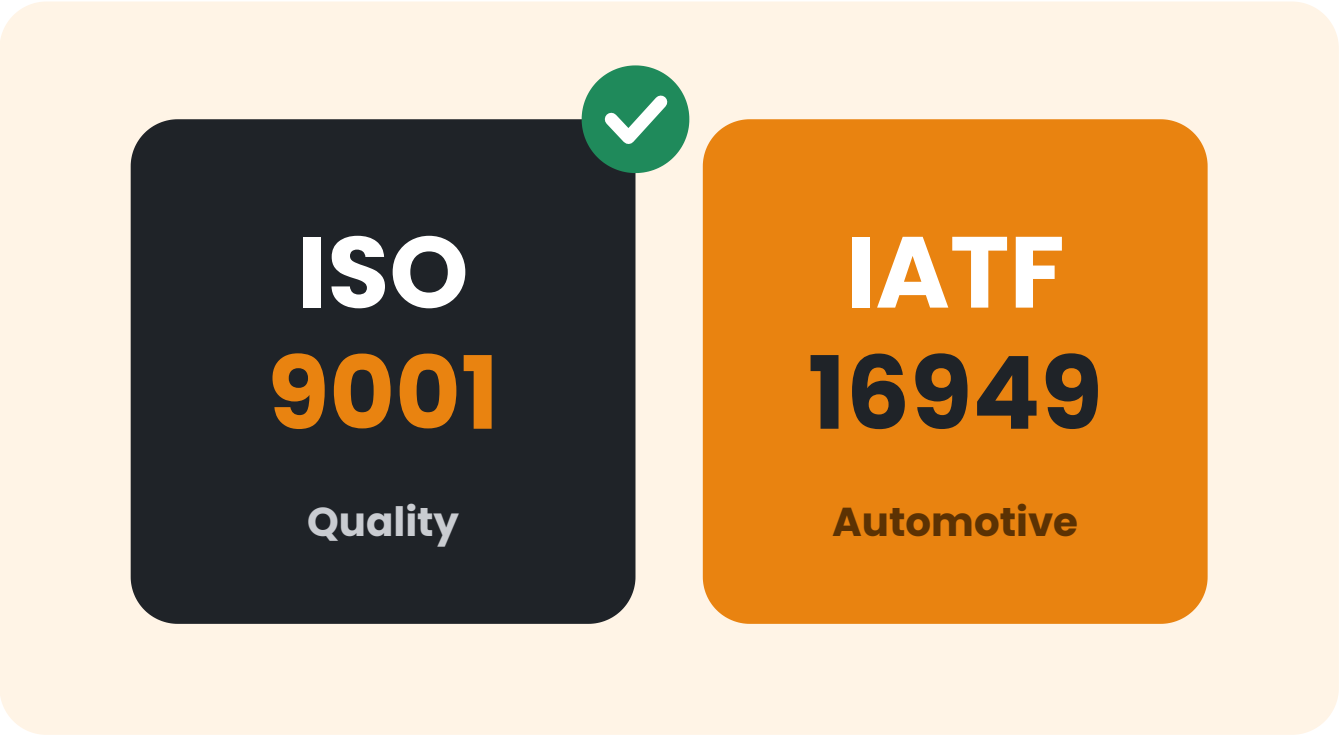
MachiNest

QUALITY BASICS

ISO 9001 & IATF 16949: Basic Training

What these quality standards are, why customers ask for them, and what they mean on the shop floor, with machine-shop examples.

Quality management | ISO 9001:2026 | IATF 16949 | Core tools | Certification



INTRODUCTION

What are ISO 9001 and IATF 16949?

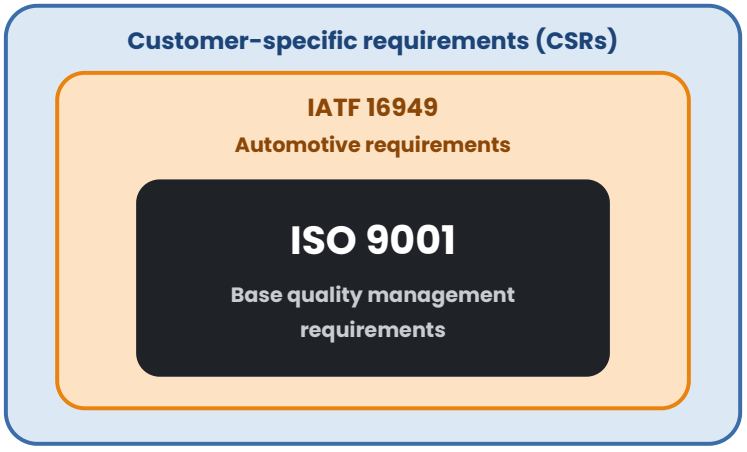
A standard is an agreed set of requirements that many companies follow, so customers can trust the result. ISO 9001 and IATF 16949 are standards for a **quality management system (QMS)**: the way a company plans, does, checks and improves its work so that every part meets the customer's needs.

ISO 9001

International standard for quality management, published by ISO (International Organization for Standardization). Any organisation can use it, of any size and in any industry: manufacturing, services, hospitals or schools.

IATF 16949

Quality management standard for the automotive supply chain, written by the IATF (International Automotive Task Force), a group of vehicle makers and their trade associations. It adds automotive requirements on top of ISO 9001 and is used together with it.



IATF 16949 is not a stand-alone standard. It includes all of ISO 9001 and adds automotive requirements. Each automotive customer then adds its own customer-specific requirements (CSRs).

Current editions (October 2026)

- **ISO 9001:2026** was published on 16 September 2026. Companies certified to ISO 9001:2015 must move to the new edition by 30 September 2029.
- **IATF 16949:2016** is the current automotive standard. A 2nd edition is in development and planned for around mid-2027.
- Always confirm dates and rules with your certification body.

WHY

Why are these standards needed?

Without a system, quality depends on a few experienced people and on luck. A QMS makes good work repeatable: the same right method, every shift, every operator, every batch.



No identification, an old drawing revision and a gauge past its calibration date: a defect is only a matter of time.



Tagged OK and reject bins, the current drawing and a calibrated gauge: the right part, made the right way.

Why customers ask for it

- Most large customers buy only from suppliers with a certified QMS.
- In automotive, vehicle makers and Tier-1 suppliers normally require IATF 16949 from their parts suppliers.
- A certificate shows that an independent auditor has checked the system.
- It is often needed to quote for export business.

Benefits

For the company

- Fewer rejects and less rework
- Less firefighting, more on-time delivery
- New customers and export orders
- Decisions based on data

For the customer

- Consistent quality, batch after batch
- Full traceability when something goes wrong
- Quick, fact-based corrective action

For our people

- Clear work instructions and training
- Less blame, more problem solving
- Skills recognised, safer workplace

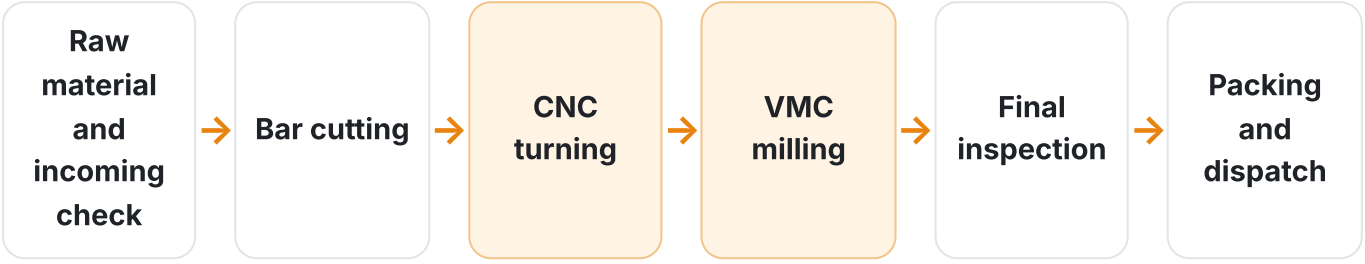
FOUNDATION

The 7 quality management principles

ISO 9001 is built on seven principles described in ISO 9000. They explain the thinking behind the requirements.

	Principle	Meaning	Shop example
1	**Customer focus**	Understand and meet customer needs	Read the drawing notes and special characteristics before setup
2	**Leadership**	Managers set direction and support quality	Owner reviews reject data every week and provides resources
3	**Engagement of people**	Everyone is competent and involved	Operators suggest kaizens and stop when something is wrong
4	**Process approach**	Manage work as connected processes	Cutting → turning → milling → inspection → dispatch, each with checks
5	**Improvement**	Keep getting better	A kaizen every month on the top rejection
6	**Evidence-based decisions**	Decide using data	Use SPC and reject data, not opinions
7	**Relationship management**	Work well with suppliers and partners	Rate raw material and heat-treatment suppliers on quality and delivery

Process approach in a machine shop



Each process has inputs, outputs, an owner, checks and records. The output of one process is the input of the next.

Easy to remember Say what you do, do what you say, prove it with records, and keep improving.

ISO 9001

How ISO 9001 is organised

Clauses 1 to 3 explain the scope and terms. The requirements are in clauses 4 to 10, which follow the Plan-Do-Check-Act (PDCA) cycle. The 2026 edition keeps this same structure.



Customer requirements come in; the PDCA cycle, led by top management, turns them into good parts and satisfied customers.

Clause	Title	In simple words
4	**Context of the organisation**	Who we are, our customers, what affects us, what the QMS covers
5	**Leadership**	Top management owns quality: policy, roles and responsibilities
6	**Planning**	Risks, opportunities, quality objectives and planned changes
7	**Support**	People, competence, machines, gauges, work environment, documents
8	**Operation**	Customer requirements, design, purchasing, production, release, nonconforming parts
9	**Performance evaluation**	Measure and analyse, internal audit, management review
10	**Improvement**	Corrective action and continual improvement

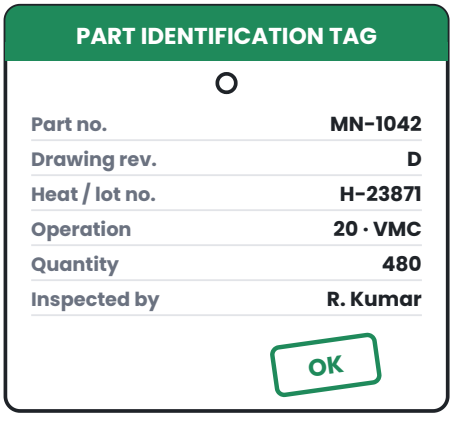
ISO 9001

ISO 9001 on the shop floor

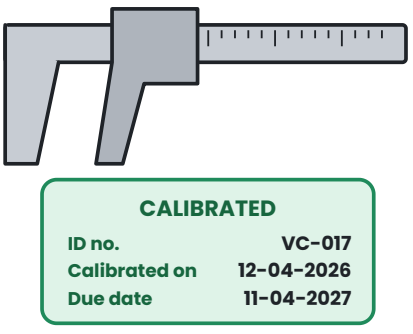
What some key requirements look like in a CNC machine shop.

Clause	Requirement	Shop example
7.1.5	Monitoring and measuring resources	Vernier, micrometer and gauges calibrated, labelled and traceable
7.2	Competence	Skill matrix for operators; training before working on a new machine
7.5	Documented information	Only the current drawing revision at the machine; records kept
8.5.1	Control of production	Setup sheet, program, work instruction and inspection plan for each part
8.5.2	Identification and traceability	Part tag with part no., heat no., operation, quantity and inspector
8.6	Release of products	Final inspection report before dispatch
8.7	Control of nonconforming outputs	Red tag and red bin for rejects; the decision is recorded
9.2	Internal audit	Trained auditors check each process as per the audit plan
10.2	Nonconformity and corrective action	Customer complaint → root cause (5 Why) → action → check it worked

Clause numbers as in ISO 9001:2015. The 2026 edition keeps the same main structure; check sub-clause numbers in your copy.



Every bin carries a tag, so any part can be traced back to its material and operation.



Every measuring instrument has an ID number and a calibration due date.

IATF 16949

What IATF 16949 adds

IATF 16949 applies to sites that make production and service parts for the automotive industry. It keeps every ISO 9001 requirement and adds requirements like these, plus each customer's own requirements (CSRs).

Clause	Topic	Shop example
4.4.1.2	**Product safety**	Extra care and approvals for safety parts such as brake or steering components
6.1.2.3	**Contingency plans**	Plan for machine breakdown, power cut or supplier failure so delivery continues
7.2.3	**Internal auditor competency**	Auditors trained in the core tools and customer requirements
8.3.3.3	**Special characteristics**	Critical dimensions marked on the drawing and control plan, checked more often
8.5.1.1	**Control plan**	For each operation: what to check, how, how often, and what to do if it is wrong
8.5.1.3	**Verification of job set-ups**	First-off approval after every setup, with a record
8.5.1.5	**Total productive maintenance**	Planned maintenance, critical spares and downtime tracking
8.5.1.6	**Management of production tooling**	Tool life, regrinding and fixture maintenance records
8.7.1.4	**Control of reworked product**	Rework only with an approved method and re-inspection
9.2.2.3	**Manufacturing process audit**	Audit every manufacturing process on the floor, across all shifts
10.2.3	**Problem solving**	A structured method such as 8D for customer complaints
10.2.4	**Error-proofing**	Poka-yoke devices listed and checked regularly
10.2.5	**Warranty management**	Analysis of parts returned from the field

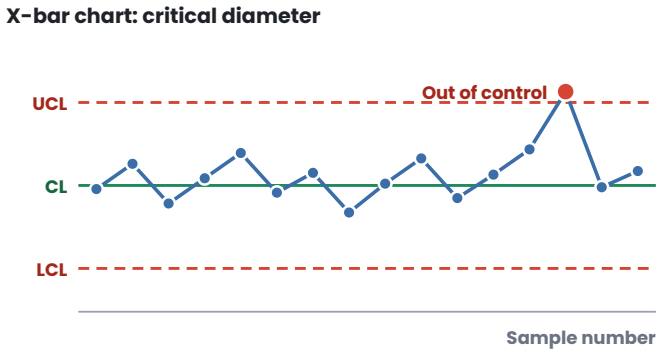
Clause numbers as in IATF 16949:2016.

IATF 16949

The automotive core tools

Automotive customers expect suppliers to use these "core tools". Most are described in AIAG manuals; FMEA follows the AIAG-VDA FMEA Handbook.

Tool	What it is	Shop example
APQP Advanced Product Quality Planning	A step-by-step plan to launch a new part without problems	New part: feasibility, process flow, FMEA, control plan, trial run, PPAP
PPAP Production Part Approval Process	Evidence to get customer approval before mass production	Dimensional report, material certificate, capability study, Part Submission Warrant (PSW)
FMEA Failure Mode and Effects Analysis	Finds what can go wrong, and how serious it is, before it happens	Risk of a wrong bore size → add an in-process gauge check
MSA Measurement System Analysis	Checks that gauges and the people measuring are reliable	Gauge R&R study on a bore gauge
SPC Statistical Process Control	Uses charts to see when a process starts to drift	X-bar/R chart and Cpk on a critical diameter
Control plan	Summary of all checks for each operation	Characteristic, method, sample size, frequency, reaction plan



A point outside the control limits means: stop, find the cause and act, before bad parts are made.

Common rules of thumb (always check your customer's requirements)

- Gauge R&R under 10%: acceptable. 10–30%: may be acceptable depending on use. Over 30%: not acceptable.
- Process capability: Cpk of 1.33 or more is a common target in production; many customers ask for 1.67 at PPAP.

SIDE BY SIDE

ISO 9001 vs IATF 16949

	ISO 9001	IATF 16949
Who uses it	Any organisation, any industry	Sites making automotive production and service parts
Written by	ISO	IATF (vehicle makers and their associations)
Stand-alone?	Yes	No, used together with ISO 9001
Customer-specific requirements	Not part of the standard	Must be met, and are audited
Core tools	Not required	Expected (APQP, PPAP, FMEA, MSA, SPC)
Who can certify	Any accredited certification body	Only certification bodies recognised by IATF
Main focus	Customer satisfaction and improvement	Defect prevention, less variation and waste in the supply chain
Current edition	ISO 9001:2026	IATF 16949:2016 (2nd edition planned ~2027)

What is new in ISO 9001:2026

- Top management must promote a quality culture and ethical behaviour
- Risks and opportunities are treated separately
- Stronger planning and control of changes
- Climate change is considered as part of the organisation's context
- Some key terms are now defined in the standard itself
- Same clause 4–10 structure: existing systems need updating, not rebuilding

CERTIFICATION

The certification journey

Certification means an independent certification body audits your system and issues a certificate. It is usually valid for three years, with surveillance audits in between.

1. **Commitment:** top management decides the scope and appoints a coordinator and team.
2. **Gap analysis:** compare current practice with the standard.
3. **Build the system:** processes, procedures, records and quality objectives.
4. **Train everyone:** awareness for all, detailed training for key people.
5. **Run it** and keep records for some months.
6. **Internal audit and management review.**
7. **Certification audit:** Stage 1 (readiness and documents), then Stage 2 (implementation on the floor).
8. **Close any findings** and receive the certificate. Then surveillance audits, and recertification every three years.

For IATF 16949

- Only a certification body recognised by IATF can certify.
- Customer-specific requirements and core tools are audited.
- The IATF certification rules set conditions before the first audit, such as a minimum period of records. Check them with your certification body.

Start from what you already do Most shops already have drawings, setup sheets, inspection reports and job cards. The standard helps you make them consistent, controlled and improving.

FOR EVERYONE ON THE FLOOR

Your role in the quality system

The system works only when people on the floor follow it every day. Every operator should be able to say "yes" to these:

1. I know the quality policy and what my job means for the customer.
2. I work from the current drawing, setup sheet and work instruction.
3. I use only calibrated gauges and check the due date before use.
4. I do a first-off check after every setup and tool change.
5. I fill in records honestly, at the time of checking.
6. I keep OK, rework and reject parts separated and tagged.
7. I stop and tell my supervisor when something is not right.
8. I suggest improvements and take part in kaizen.

Myths and facts

Myth	Fact
"ISO means a lot of paperwork."	Keep the documents you need to work right and to prove it. Remove the rest.
"Quality is the QC department's job."	Quality is made at the machine. QC checks and supports.
"Once certified, the job is done."	Surveillance audits check every year that the system is used and improving.
"IATF is only for big companies."	Any supplier of automotive production parts can be asked for it, including small shops.
"The auditor wants to catch people."	The auditor checks the system. Findings show where it can improve.

TEST YOURSELF

Quick self-check

1. **ISO 9001 can be used by:**
   (a) Only factories (b) Any organisation (c) Only automotive companies
2. **IATF 16949 is used:**
   (a) Together with ISO 9001 (b) On its own (c) Instead of drawings
3. **Which clause covers internal audit and management review?**
   (a) 7 (b) 8 (c) 9
4. **A gauge past its calibration due date should be:**
   (a) Used carefully (b) Taken out of use and sent for calibration (c) Used only for rough parts
5. **PPAP is used to:**
   (a) Get customer approval before mass production (b) Pay suppliers (c) Plan holidays
6. **SPC helps you see:**
   (a) The machine price (b) Operator attendance (c) When a process is drifting
7. **After every setup you should:**
   (a) Start mass production (b) Do a first-off check (c) Wait for the customer
8. **ISO 9001:2015 certificates must move to the 2026 edition by:**
   (a) 30 September 2029 (b) 2027 (c) Never

Answers: 1-b, 2-a, 3-c, 4-b, 5-a, 6-c, 7-b, 8-a

Glossary

Term	Meaning
QMS	Quality management system: how a company manages quality
NC	Nonconformity: something not meeting a requirement
CAPA	Corrective and preventive action
CSR	Customer-specific requirement
PSW	Part Submission Warrant, the PPAP summary form
Cpk	Process capability index: how well a process fits the tolerance
8D	Eight-discipline problem-solving method
Calibration	Comparing an instrument with a known standard
Traceability	Being able to trace a part back to its material, process and people